

# Austin Tan

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## EDUCATION

### New York University (NYU)

New York City, NY

*Bachelor of Science in Computer Science (GPA: 3.92/4.0)*

*Expected May 2028*

- **Relevant Coursework:** Data Structures and Algorithms, Design and Analysis of Algorithms, Operating Systems, Object-Oriented Programming, Machine Learning, Artificial Intelligence, Computer Architecture, Databases, Probability & Statistics, Discrete Mathematics
- **Activities:** Tech@NYU Dev Team - Spring 2025

## TECHNICAL SKILLS

**Languages:** C/C++, Python, Go, Java, TypeScript, JavaScript, SQL, Bash

**Frameworks & Tools:** React, Next.js, Node.js, CMake, GoogleTest, Git, Jenkins, CI/CD, RAG

**Systems & Infrastructure:** Linux/Unix, TCP/UDP, Docker, Kubernetes, PostgreSQL, MySQL, Redis, GCP, AWS

## WORK EXPERIENCE

### Amazon Web Services (AWS)

Arlington, VA

*Software Development Engineer Intern*

*Sep 2026 - Present*

### New York University

New York City, NY

*Undergraduate Research Assistant - Distributed Systems & Networking*

*Apr 2026 - Present*

- Designed and implemented a wide area network (WAN) consensus protocol in Go that uses clock-synchronized predictive scheduling to eliminate per-request inter-replica coordination from the common path, enabling commits in a single WAN RTT
- Evaluated the protocol against Paxos, EPaxos, SwiftPaxos, and CURP across geo-distributed GCP and bare-metal CloudLab clusters, sustaining 200K+ requests/sec and pushing saturation several-fold beyond the strongest baseline while maintaining comparable latency
- Built automated multi-region benchmarking and custom TCP networking infrastructure to evaluate throughput, latency, and protocol behavior under controlled WAN conditions
- **Selected Research Acceptance(s):** SOSP SysDW '26

### MathWorks

Natick, MA

*Software Engineer Intern - MATLAB Language QE Team*

*May 2026 - Aug 2026*

- Migrated 5 core MATLAB language test suites from a deprecated test harness to the modern xUnit-based MATLAB Unit Testing Framework, eliminating legacy test infrastructure and reducing long-term technical debt
- Owned functional validation for dynamic updates to MATLAB built-in classes, a feature required for the R2027a release; implemented C++ built-in classes/functions and MATLAB tests covering inheritance, shadowing, overriding, invalid user input, and error behavior
- Validated end-to-end feature semantics to ensure downstream MATLAB users receive both correct runtime behavior and appropriate errors during execution

### CrowdVolt (YC W24)

New York City, NY

*Software Engineer Intern*

*Oct 2025 - Apr 2026*

- Shipped a full-stack price alert system used 4,000+ times across the marketplace, owning the PostgreSQL data model, REST APIs, Next.js/React frontend, deployment, and production monitoring end-to-end
- Developed an LLM-based data normalization pipeline that processed 1,000+ event requests across artist, location, and date fields, automating manual triage and surfacing structured results through an internal operations dashboard
- Built a Redis-backed recovery and triage service for a multi-stage bank verification flow, helping unblock stalled seller verifications and improving bank-linking conversion
- Implemented an internal operations platform used daily by 15+ employees for project management, shift scheduling, analytics, transaction monitoring, and cart validation, helping reduce marketplace manipulation to near zero

## SELECTED PROJECTS

### Distributed Storage System | C++, liburing, Linux, Docker, CMake, GoogleTest

- Built a high-performance distributed append only log in C++20 with a custom UDP protocol and an asynchronous persistence layer with io\_uring, including checksum validation, crash recovery, and thread-safe concurrent writes
- Implemented a simplified version of Network-Ordered Paxos (NOPaxos) with sequencer based replication to eliminate cross replica coordination along with failure detection and automatic failover; validated correctness through extensive unit and concurrency tests

### Personalized Music Recommendation System | Python, NumPy, Pandas, scikit-learn, Matplotlib, Seaborn, Gradio

- Built a hybrid music recommendation engine combining SVD-based collaborative filtering and KNN, improving Precision@5 by 150% (0.036 → 0.091) while reducing training MSE by 60% (1.08 → 0.43)
- Trained and evaluated the system on 166K+ Last.fm interactions across 76K+ songs, matching 114K+ Spotify tracks for feature enrichment and benchmarking SVD, KNN, and hybrid approaches using Precision, Recall, and NDCG@k

## PERSONAL INTERESTS

**Hobbies:** Weightlifting (225lb Bench Press), Classical Piano (Chopin), Backcountry Hiking (Appalachian Trail), Saltwater Fishing